The way it works is this: when the universe reaches a state of steady exponential expansion, you can define a radius (from wherever you are) beyond which the rest of the cosmos is forever hidden. It's a true horizon in the sense that nothing beyond it could ever reach you. It turns out that this horizon, like a black hole's horizon, also has an entropy associated with it, and thus a temperature. The difference is that instead of the heat going *out* like it does with a black hole, it goes *in*. The temperature is very small—something like 10^{-40} degrees above absolute zero—but when everything else has decayed, this radiation is all that's left to contain all the entropy of the universe. When the universe gets to this pure de Sitter state, it is a *maximum entropy universe*. From that point on, there is no way for the universe's total entropy to increase, which means, in a very real sense, the arrow of time is … gone.

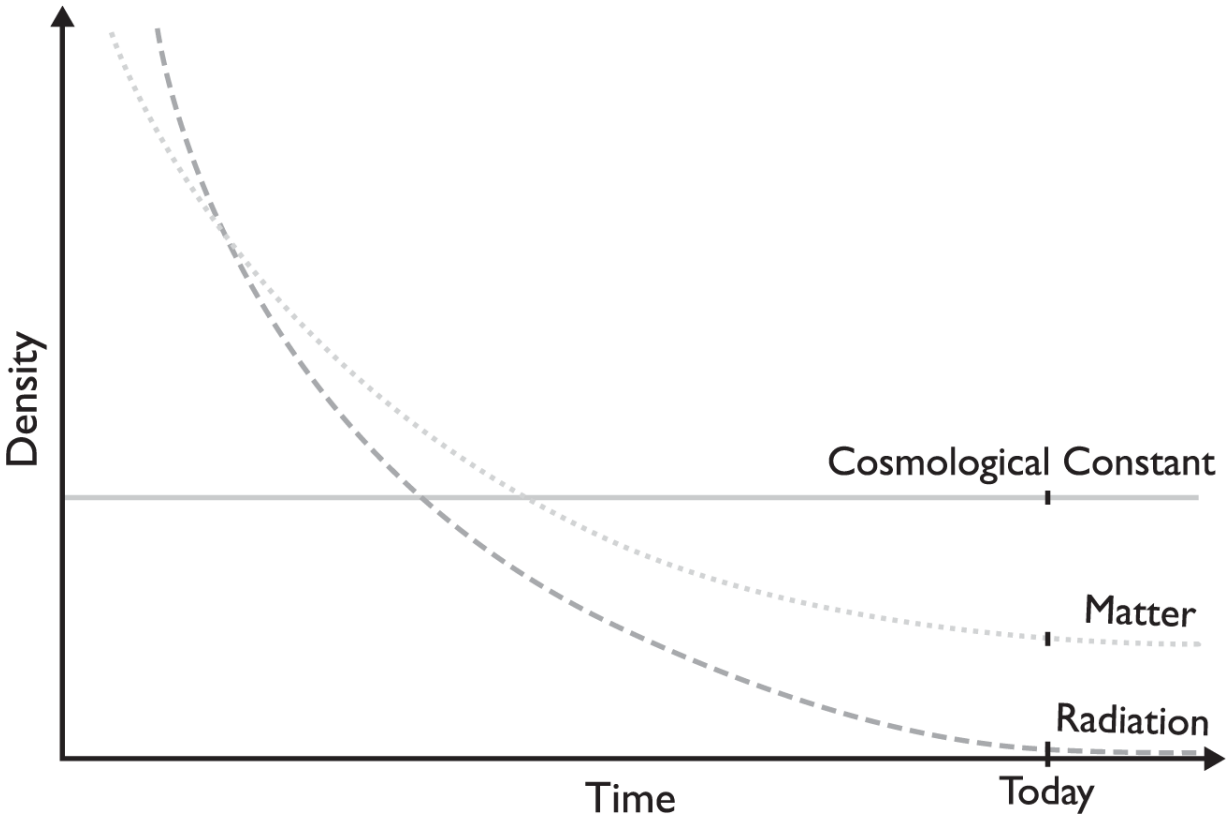


Figure 13: Density of matter, radiation, and cosmological constant over time. Because the density of dark energy (in the form of a cosmological constant) doesn't change as the universe expands, while everything else dilutes, it comes to dominate the energy density of the universe. Today, dark energy makes up around 70 percent of the universe, while matter is around 30 percent and radiation is a tiny amount.